MILITARY SPECIFICATION

CARTRIDGE, FUZELESS, 25MM, HIGH EXPLOSIVE INCENDIARY TRACER, (HEI-T), M792

This specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for one type of percussion primed cartridge, Fuzeless, 25mm, High Explosive Incendiary Tracer, M792 for use in 25mm guns (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein.Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions, and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

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MIL-STD-644	Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, packing and Marking of Small Arms Ammunition
MIL-STD-651	Visual Inspection Standards for 20mm Ammunition and Components
MIL-STD-1168	Lot Numbering of Ammunition
MIL-STD-1167	Ammunition Data Card

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER (ARDEC)

9391400	Interface Control Drawing, Maximum Profile and Alignment, Cartridge M792, M793, and HPT
12597642	Cartridge, Fuzeless, 25mm, High Explosive Incendiary Tracer, M792
12013674	Shipping Container-Unlinked 25mm Ammunition (80 Rds)
AS12013566	25mm Ammunition Ballistic Test Methods
AS12013734	Ammunition, 25mm, Packaging, Packing and Marking

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification, shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless specfic exemption has been obtained.

3. REQUIREMENTS

3.1 General. The fuzeless cartridge shall be in accordance with Drawing 12597642, referenced specifications, publications and other requirements specified herein.

a. Material. Materials shall be in accordance with applicable drawings and specifications.

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b. Manufacturing process. All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without prior approval of the contracting officer (see 6.7).

3.1.1 Cartridge. The fuzeless cartridge shall comply with all requirements specified on Drawing 12597642, all associated drawings, and with all requirements specified in applicable specifications.

3.2 Projectile extraction. The tensile force necessary to extract the projectile from the sample cartridge case shall be as specified by Drawing 12597642 (see 4.4.1).

3.3 Projectile torque. The sample projectile shall withstand a minimum of 30 in-lbs torque without rotational movement (see 4.4.3).

3.4 Velocity. The average muzzle velocity of the sample cartridges, conditioned shall be 1100 ± 20 meters per second (mps). The sample standard deviation shall not exceed 5 mps (see 4.4.4).

3.5 Pressure. The average peak chamber pressure of the sample cartridges, conditioned at 21° ± 3°C, shall not exceed 402 megapascals (MPa). The average peak chamber pressure of the cartridges, when functioning at any individual temperature from -46°C to 62°C, plus five standard deviations of peak chamber pressure, shall not exceed 496 MPa and no individual sample cartridge shall exceed 496 MPa. Pressure shall be measured with a piezoelectric type pressure transducer (see 4.4.4).

3.6 Dispersion. The standard deviation of impacts in both the horizontal and vertical directions of the sample fuzed projectiles shall not exceed 0.50 milliradians when the cartridges are conditioned at 21° ± 3°C (see 4.4.5). For this test, the test barrel will be fitted and fired with a muzzle brake.

3.7 Action time. The action time of the sample cartridges shall not exceed 6.0 ms at any cartridge temperature from -46°C to 62°C (see 4.4.6).

3.8 Projectile trace. The sample projectile shall exhibit a visible trace when viewed against a dark background, for a minimum of 3.5 seconds from the gun muzzle flash at any cartridge temperature form -46°C to 62°C (see4.4.7).

3.9 Projectile function. See 4.4.8.

3.9.1 Projectile impact function. The sample fuzed projectiles shall function upon impact or up to one meter behind a target of 0.063 inch nominal thickness aluminum plate 2024-T3 at 0° obliquity, at a distance of 200 ± 5 meters from the weapon at any temperature from -46°C to 62°C.

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3.9.2 Projectile self-destruct function. The sample fuzed projectile shall function between 6.2 seconds minimum and prior to impact at any cartridge temperature from -46°C to 62°C.

3.10 Function and casualty. The sample cartridges, at 21° ± 10°C shall function without firing defects and casualties (see 4.4.9).

3.11 Workmanship. All parts and assemblies shall be fabricated, loaded, and assembled in a thorough workmanship manner. They shall be free of burrs, sharp edges, cracks, scratches, dents, folds, wrinkles, buckles, dirt, grease, oil, rust, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and clearly defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Quality assurance terms and definitions shall be in accordance with MIL-STD-109 and the provisions of MIL-A-2550 and MIL-A-48078 apply.

4.1.1 Responsibility for compliance.All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections.The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3)

b. Lot acceptance testing

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4.3 Quality conformance inspection.

4.3.1 Component parts. Unless otherwise specified, component parts shall be homogeneous and of a quantity convenient to the contractor, except that each cartridge lot shall contain:

a. Cartridge cases from one unchanged process, from one lot inter fix number, and one manufacturer.

b. projectiles from one unchanged process, from one lot inter fix number, and from one manufacturer.

c. Primers from one lot, from one lot interfix number, and one manufacturer.

d. Propellant from one lot and one manufacturer.

e. Explosive fill from one lot and one manufacturer.

f. Hybrid lots are not permitted. Each fuzeless carrier lot will be assembled from a single case, projectile, primer, and propellant lot.

4.3.1.2. Assembled cartridge lot. Each lot of ammunition shall be identified as to type, caliber, and model, as welll as a lot number in accordance with MIL-STD-1168, supplemented as directed by the procuring activity. Each packed lot of ammunition shall further be identified by a national stock number assigned by the procuring activity.

4.3.2 Examination for defects. See MIL-A-48078. MIL-STD-651 shall apply in defining and evaluating cartridge visual defects as applicable.

a. Critical defects. Unless otherwise specified, one hundred percent examination shall be performed for all critical defects.If a visual critical defect is found in a sample either just prior to a firing test, or after a firing test (and the defect is not due to the firing), the lot shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects. If a critical defect is found during packing, the portion of the lot remaining to be packed shall be rejected. The lot shall then be rescreened and resubmitted for visual inspection of critical defects.

b. Major and minor defects. Inspections shall be performed on an individual basis in accordance with classification of defects using applicable sampling plans and acceptance criteria of MIL-STD-105. All nonconforming cartridges shall be rejected. The following AQL’s are assigned to major and minor defects:

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Individual basis

Major	0.25%
Minor	0.40%

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70690 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.3.1	Charged Body, Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	HEI Charge depth (depth applies to entire face of fuze well)		100%	3.2	Gage
2.	Explosive in threads		100%		Visual
3.	Crack or split in projectile		100%		Visual
Major					
101.	Improper trace closure		.25		Visual/Gage
Minor					
201.	Workmanship		.40	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70690 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.3.2	Case, Cartridge, Primed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Inverted or missing primer or ignitor		100%		Visual
Major					
101.	Depth of primer seating		.25		
Minor					
201.	Waterproof missing around primer		.40		Visual
202.	Primer crimp missing		.40		Visual
203.	Workmanship		.40	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-70690 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 12597642
4.3.3.3	Cartridge		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Split or perforated case		100%		Visual
2.	Crimp missing (case/projectile)		100%		Visual
3.	Low propellant charge		100%		Gage
Major					
101.	Cocked, loose, missing or inverted primer		.25		Manual/Visual
102.	Overall length, max.		.25		Gage
103.	Improper or incomplete crimp (case/projectile)		.25		Visual
104.	Profile and alignment, max.		.25		9391400 1/ 2/
105.	Corrosion		.25		Visual
106.	Head configuration improper		.25		Visual
107.	Metal slivers on head face		.25		Visual
108.	Nose plug not seated		.25		Visual
109.	Rotating band damaged		.25		Visual

NOTES: 1/ A dead weight load of 60 pounds max may be ued to insert the cartridge into the gage.
2/ Internal configuration only. Submit final design in accordance with 6.5.

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QUALITY CONFORMANCE INSPECTION

MIL-C-70690 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 12597642
4.3.3.3	Cartridge				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201.	Marking incorrect, incomplete, illegible or missing		.40		Visual
202.	Scratch, dent, buckle, bulge, wrinkle or fold in case		.40		Visual
203.	Incorrect type cartridge or components		.40		Visual
204.	Workmanship		.40	3.11	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.3.4 Testing. The tests listed in Table I shall be performed on each cartridge lot in accordance with the test methods prescribed in section 4.4. Unless otherwise indicated, tests shall be conducted with samples at 20°C ± 10°C. Sample size and acceptance criteria for each test shall be as specified. only cartridges having met the visual and dimensional requirements and having been selected in such a manner that the sample is representative of the entire lot shall be used in the tests. The selected cartridges shall be thoroughly mixed before being divided into samples for the various tests. The combining of tests is permitted (see 6.4).

4.3.4.1 Firing defects and casualties. Firing defects and associated criteria shall be as specified in Table II. For the defect definitions, see AS12013566.

4.3.4.2 Unlisted firing defects. The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test, indicates that the product may be unsuitable for the purpose intended.

4.3.5 Packaging, packing and marking inspection. MIL-STD-644 shall be used as a guide for inspection of packaging, packing and marking as applicable to Dwg. 12013674.

4.3.6 Inspection equipment. Inspection equipment required to perform examinations and tests prescribed herein shall conform to the provisions of MIL-A-48078.

4.4 Methods of inspection.

4.4.1 Projectile extraction. The method shall be as prescribed in AS12023566.

4.4.2 Propellant contamination. The propellant from each of the cartridges will be examined visually and chemically for contamination, as prescribed in AS12013566.

4.4.3 Projectile torque. Each cartridge of the test sample shall be marked axially across the rotating band onto the case neck. Torque, to the specified requirement, shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case detectable by misalignment in the scratch mark, shall be recorded.

4.4.4 Velocity and pressure. The methods of test shall be as prescribed in AS12013566. Statistics necessary for testing of results against acceptable criteria shall be computed (see 6.3). A correction factor of 0.46 mps per meter shall be applied to the recorded velocity at the measured range to obtain muzzle velocity.

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Table I. Lot acceptance testing

CLASSIFICATION OF DEFECTS & TESTS MIL-C-70690 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
	Cartridge, Fuzeless, 25mm HEI-T, M792 1/				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile extraction 2/	32	0-1	3.2	4.4.1
	Propellant contamination 3/	32	0-1	3.11	4.4.2
	Projectile torque 4/	32	0-1	3.3	4.4.3
	Velocity 5/ 12/	45	0-1	3.4	4.4.4
	Pressure 6/		0-1	3.5	4.4.4
	-46°C ± 3°C	45			
	21°C ± 3°C	45			
	62°C ± 3°C	45			
	Dispersion 7/ 12/	60		3.6	4.4.5
	Action time 8/		0-1	3.7	4.4.6
	-46°C ± 3°C	45			
	21°C ± 3°C	45			
	62°C ± 3°C	45			
	Projectile Trace 9/			3.8	4.4.7
	-46°C ± 3°C	39	2-7		
		120	6-7		
	21°C ± 3°C	39	2-7		
		120	6-7		
	62°C ± 3°C	39	2-7		
		120	6-7		

NOTES:

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Table I. Lot acceptance testing

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE Cartridge, Fuzeless, 25mm HEI-T, M792 1/		SHEET 2 OF 2		DRAWING NUMBER NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Projectile function 10/			3.9	4.4.8
	Impact function			3.9.1	4.4.8
	-46°C ± 3°C	57	1-4		
		114	3-4		
	21°C ± 3°C	57	1-4		
		114	3-4		
	62°C ± 3°C	57	1-4		
		114	3-4		
	Self destruct			3.9.2	4.4.8
	-46°C ± 3°C	39	1-6		
		120	5-6		
	21°C ± 3°C	39	1-6		
		120	5-6		
	62°C ± 3°C	39	1-6		
		120	5-6		
	Function and casualty 11/ 12/	200		3.10	4.4.9

NOTES:

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

NOTES:

1/ The lot shall be rejected if in any firing test one or more projectiles remain in the bore or if the number of firing defects exceeds any of the acceptance criteria of Table II. Accept or reject criteria for firing casualties are shown in Table II. Except as otherwise specified and in the function and casualty test where the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristics being tested, an additional shot shall be fired.

2/ Projectile extraction. Failure of one or more sample cartridges to comply with the minimum requirements shall cause rejection of the lot.

3/ Propellant contamination. The cartridges used for projectile extraction shall be used to inspect the propellant. The presence of foreign matter in the propellant of any of the test cartridges will cause rejection of the lot (see 3.11).

4/ Projectile torque. Failure of one or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot.

5/ Velocity. If the sample fails to comply with the requirements, the lot shall be rejected.

6/ Pressure.

a. +20°C: If the sample fails to comply with the requirement, the lot shall be rejected.

b. -46° and +62°C: If the average peak chamber pressure of a 45 cartridge sample plus two standard deviations exceeds 496 MPa, or it an individual sample cartridge exceeds 496 MPa, the lot shall be rejected.

7/ Dispersion. Six (6) fuzed, ten round groups shall be fired for dispersion. No discards will be allowed. The six groups shall be tested for homogeneity. Should this test show that all groups are homogeneous, then the groups may be pooled. If the dispersion of the pooled sample cartridges exceeds the applicable requirement, the lot shall be rejected. Should the test for homogeneity show that any one group is not homogeneous, the lot shall be rejected.

8/ Action time. If any sample cartridge fails to meet the requirements for action time, the lot shall be rejected.

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9/ Projectile trace. The fuzed cartridges used for self-destruct shall be used for the projectile trace test. The cartridges must meet the minimum requirement for trace. If more than two but less than seven cartridges of the sample fails to comply, a second sample of 81 cartridges shall be tested. The lot shall be rejected if seven or more cartridges of the combined sample fail to comply with the applicable requirement.

10/ Projectile function.

a. Impact function. The fuzed cartridges must meet the minimum requirement for impact. If more than one but less than four cartridges of the sample fails to comply, a second sample of 57 cartridges shall be tested. The lot shall be rejected if four or more cartridges of the combined sample fail to comply with the applicable requirement.

b. Self-destruct. The fuzed cartridges must meet the minimum requirement for self-destruct. If more than one but less than six cartridges of the sample fail to comply, a second sample of 81 cartridges shall be tested. The lot shalll be rejected if seven or more cartridges of the combined sample fail to comply with the applicable requirement.

11/ Function and casualty. The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable reject number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable accept number, but less than the applicable reject number, a second sample of 400 cartridges shall be selected. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in nubers equal to or greater than the applicable reject number.

12/ Ammunition conditioning. Ammunition shall be conditioned at 21° ± 3°C.

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Table II. Firing defects.

	Acc	Rej
Cartridge:		
Misfire	0	2
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
Premature projectile function	0	1
Primer:		
Primer leak	9	25
Primer perforation	1	6
Loose primer	0	1
Blown primer	0	1
Case:		
Longitudinal split 1/		
H or S	10	31
G or J	4	11
K, L or M	0	1
Circumferential rupture, partial 1/		
S, J or K	4	11
G or L	2	7
Circumferential rupture (complete) 1/		
Detached metal 2/	0	1
Projectile:		
Rotating band separation 3/		
Complete or partial separation	0	3
Metal parts separation 4/	0	1

NOTES:

1/ See Figure 1 for classifying splits and rupture in fired cartridge cases. If a longitudinal split of circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classified as a complete circumferential rupture.

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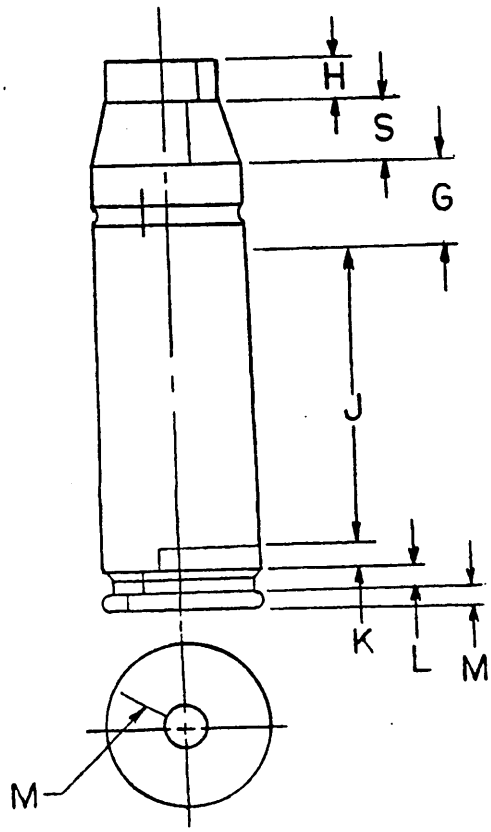
2/ Metal sheared or missing-from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck in the crimped area.

3/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing.

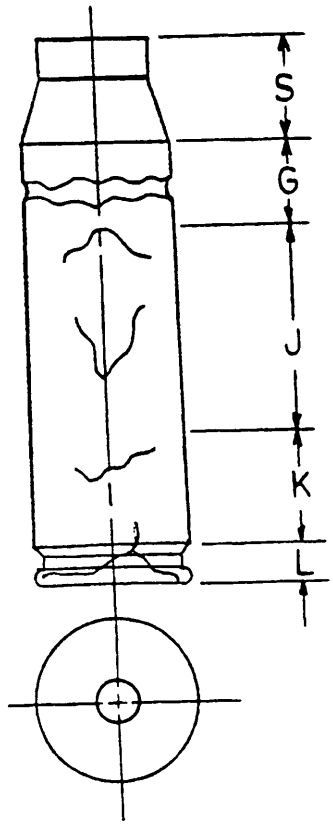
4/ Separation or breakup of projectile part(s), as evidenced by recovering of the part(s) or fragment(s) or hole(s) in the fragmentation screen(s), shall be classed as a defect. The lot shall not be penalized for evidence of any individual fragment with a weight of less than 0.10 grams.

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Splits



Ruptures



G-IN LINK GROOVE SECTION

H-IN MOUTH OR NECK

J-IN MAIN BODY SECTION, NO CLOSER THAN 1/4" TO EXTRACTOR GROOVE

K-IN LOWER BODY SECTION FROM FORWARD EDGE OF EXTRACTOR GROOVE

L-IN EXTRACTOR GROOVE

M-IN CASE HEAD

S-IN SHOULDER SECTION

G-IN LINK GROOVE SECTION

S-IN NECK AND SHOULDER

J-IN UPPER 3/4 BODY SECTION

K-IN LOWER 1/4 BODY SECTION ABOVE EXTRACTOR GROOVE

L-IN HEAD AND EXTRACTION GROOVE

FIGURE 1. Classification of defects for case, cartridge; 25mm

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4.4.5 Dispersion. The method of test and measurement of targets shall be as prescribed in AS102013566 and in Table I. The test range shall be 100 meters.

4.4.6 Action time. The method of test shall be as prescribed in AS12013566.

4.4.7 Projectile trace. The method of test shall be as prescribed in AS12013566. Visual observation(s) or photographic witness of trace characteristics shall be made from suitable position(s).

4.4.8 Projectile function. The method of test shall be as prescribed in AS12013566. The aluminum target shall be located as required from the gun muzzle normal to the line of fire. Observation of projectile functioning shall be made from a suitable position.

4.4.9 Function and casualty. The method of test shall be as prescribed in AS12013566. The test sample shall be assembled in belts and fired in bursts of 25 + 5 rounds in a 25mm gun at a rate of 200 + 50 rounds per minute. The weapon barrel shall be at ambient temperature at beginning of the test and cooled to ambient after each 100 rounds (see Table II).

4.4.10 Defective weapon. The test barrel shall be inspected prior to testing. No test barrel which has had more than 5000 rounds fired through it should be used.

4.5 In-process controls.

4.5.1 Uniformity of charge. A machine shall be chosen which is known to drop charges with better than average uniformity and which gives constant results in the daily production tests. If relative machine performance is unknown, special tests should be fired to assure selection of a good machine.

4.5.2 Loading control. A quantity of 20 rounds shall be fired for pressure and velocity for each four hours of and eight hour shift to establish velocity and pressure values, and check the charge weight and uniformity of charge weight every hour. Once established, the charge weight shall remain constant throughout loading.

5 PACKAGING.

5.1 Packing level A. The type of ammunition pack shall be in accordance with instructions from the procuring agency. Ammunition shall be packed in accordance with Dwg. 12013674.

5.2 Marking. Marking on the packed container shall be in accordance with Dwg. 12013674, as applicable.

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6. NOTES

6.1 Intended use. The fuzeless cartridges covered by this specification are intended for use in single shot test barrels for M758 Fuze lot acceptance and testing.

6.2 Ordering data. See MIL-A-48078.

6.3 Computations. Standard deviation. Where computation of a sample standard deviation is specified for determination of lot acceptance, the method of computation will be:

$$S = \sqrt{\frac{\Sigma(X_i - \bar{X})^2}{(n - 1)}}$$

Where: X_i = each individual value
$\bar{X}$ = sample arithmetic mean
n = sample size

6.4 Combining tests. In order to minimize testing costs, tests may be performed concurrently on the sample cartridges provided that the test results are not affected by this procedure. Component level testing cannot be substituted for cartridge level requirements unless approved by the PCO.

6.5 Submission of inspection equipment designs for approval. Submit equipment designs as reqireed to Commnder, ARDEC ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.6 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the contracting officer, the contracter should forward requested records of examinations or tests to Commander ARDEC ATTN: AMSMC-QAF-S (D), Picatinny Arsenal, NJ 07806-50000 This address will be specified on the Contract Data Requirements Lists, DD Form 1423 in the contract. Unless otherwisse specified, data item DI-R-1714 will apply.

6.7 Process deviation. Aprocess deviation is defined as a change in the approved basic method of manufacture, or an operational change which alters the metallurgical or physical properties of the item.

6.8 AQL's. The optional use of AQL values for classes of defects, with individual major defect limitation, is intended to assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

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6.9 Premature functioning. Is defined as the occurrence of any of the following:

a. Any function within the weapon.

b. Any function within one meter of the weapon.

c. Any function within 10 meters of uninterrupted flight from the weapon.

6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identifed as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, US Army Armament Research and Development Command (ARRADCOM) or US Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.11 Subject term (key word) listing.

Cartridge, Fuzeless, 25mm, High Explosive Incendiary Tracer
Small Arms Ammunition

Custodian:
Army-AR

Preparing activity
Army-AR

(Project 1305 - AC90)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

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MIL-C-70690 (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-C-70690(AR)
NOTICE 1
29 November 1995

MILITARY SPECIFICATION

CARTRIDGE FUZELESS, 25MM, HIGH EXPLOSIVE INCENDARY TRACER, (HEI-T), M792

This notice should be filed in front of MIL-C-70690(AR), dated 16 September 1988.

MIL-C-70690(AR) dated 16 September 1988 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity:
Army-AR

AMSC N/A

FSC 1305

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